

Generation Interconnection System Impact Study Report

For Queue Project Q19-1173

Solway 138 kV
October 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1173. The Interconnection Customer proposes a 157.3 MW wind facility interconnecting at the Solway 138 kV substation in the Maryland area. The request was received on March 23, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 376 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,833,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Leander - Solway 138 kV	Sheffield Wind - Swansea 138 kV	149.4	211	100.0
Petersburg - Leander 138 kV	Sheffield Wind - Swansea 138 kV line	140.6	215	100.0
Solway - S. Macon 138 kV Line	Bixby Hill - Tombstone 500 kV	134.9	204	38.8
Deer Valley - Chimayo 138 kV	Aztec 345-138 kV XFMR	131.2	172	32.2
Port Hudson - Chimayo 138 kV	Aztec 345-138 kV Autotransformer	128.5	184	32.2
Aztec - Port Hudson 138 kV	Aztec 345-138 kV Transformer Bank	127.0	206	32.2
S. Macon - Clay 138 kV	Tombstone - Bixby Hill 500 kV	120.8	213	38.8
Neevin - Petersburg 138 kV	Sheffield Wind Substation - Swansea 138 kV L	119.1	243	100.0
Dolphus M Grainger - Aztec 138 kV	Tombstone - Bixby Hill 500 kV	118.2	340	38.8
Dolphus M Grainger - Sheffield Wind Substation 138 kV line	Bixby Hill Substation - Tombstone 500 kV	117.2	351	38.8
Sheffield Wind - Swansea 138 kV	Tombstone - Bixby Hill 500 kV	114.5	387	38.8
Clay Substation - Swansea 138 kV	Tombstone - Bixby Hill 500 kV	111.4	223	38.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Leander - Solway 138 kV	Rebuild line with higher-capacity conductor	\$1,729,000	41% (\$702,000)
Petersburg - Leander 138 kV	Rebuild line with higher-capacity conductor	\$18,442,000	41% (\$7,490,000)
Solway - S. Macon 138 kV Line	Rebuild line with higher-capacity conductor	\$10,169,000	46% (\$4,648,000)
Deer Vly - Cimayo Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,370,000	28% (\$940,000)
Port Hudson - Chimayo 138 kV	Rebuild line with higher-capacity conductor	\$3,154,000	28% (\$880,000)
Port Hudson - Aztec 138 kV	Rebuild line with higher-capacity conductor	\$1,761,000	28% (\$491,000)
Clay - S. Macon 138 kV	Rebuild line with higher-capacity conductor	\$6,328,000	46% (\$2,893,000)
Neevin - Petersburg 138 kV	Rebuild line with higher-capacity conductor	\$4,839,000	41% (\$1,965,000)
Dolphus M Grainger - Aztec 138 kV	Rebuild line with higher-capacity conductor	\$2,760,000	28% (\$772,000)
Dolphus M Grainger - Sheffield Wind Substation 138 kV line	Rebuild line with higher-capacity conductor	\$3,555,000	28% (\$994,000)
Sheffield Wind - Swansea 138 kV	Rebuild line with higher-capacity conductor	\$1,731,000	28% (\$484,000)
Clay Substation - Swansea 138 kV	Rebuild line with higher-capacity conductor	\$3,445,000	46% (\$1,574,000)
	Total allocated to project		\$23,833,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.